Stack Application

Data Structures and its Applications

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Quiz 1: Stack Application

How do we make out that (()))) is not well formed in our algorithm?

a) By counting and comparing the number of left and the number of right parentheses

b) By encountering popping from an empty stack

c) By having a non-empty stack when the input string has been completed processed

d) None of these

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Quiz 2: Stack Application

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Quiz 3: Stack Application

On encountering a left parenthesis,

a) Pop the top element from the stack

b) Push some marker onto the stack

c) Should push) only on the stack

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Quiz 4: Stack Application

When an expression with 3 different type of brackets is to be checked for well formedness,

a) We require 6 different types of symbols to put on the stack

b) We require one symbol to put on the stack

c) We require 3 different types of symbols to put on the stack

d) None of these

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Quiz 4: Stack Application

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Quiz 5: Stack Application

If at some point in processing a bracketed expression, if the stack is empty

a) It means the expression is not well formed

b) It means the expression has had no parentheses until this point

c) It definitely means the expression is well formed no matter what follows.

d) None of these

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Quiz 5: Stack Application

If at some point in processing a bracketed expression, if the stack is empty

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Stack : Applications

stack

- All stack operations happen at the stack top.
- Push adds an element to the top.
- Pop removes the top element.
- Peek inspects the top element.
- The implementation may use an array or a linked list
- The interface is what the user needs to know.

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Stack : Applications

Stack : Well formed parentheses

- Why Counting Parentheses Is Not Enough
- Examples: (3 + 4) and ((5 + 6) * (7 + 8)) are well formed.
- Ignoring other characters gives () and (()()).
- Equal numbers of '(' and ')' are necessary, but not sufficient.
- Example:)(is invalid because a right parenthesis occurs before a left one.
- Example: ())(()) is also invalid

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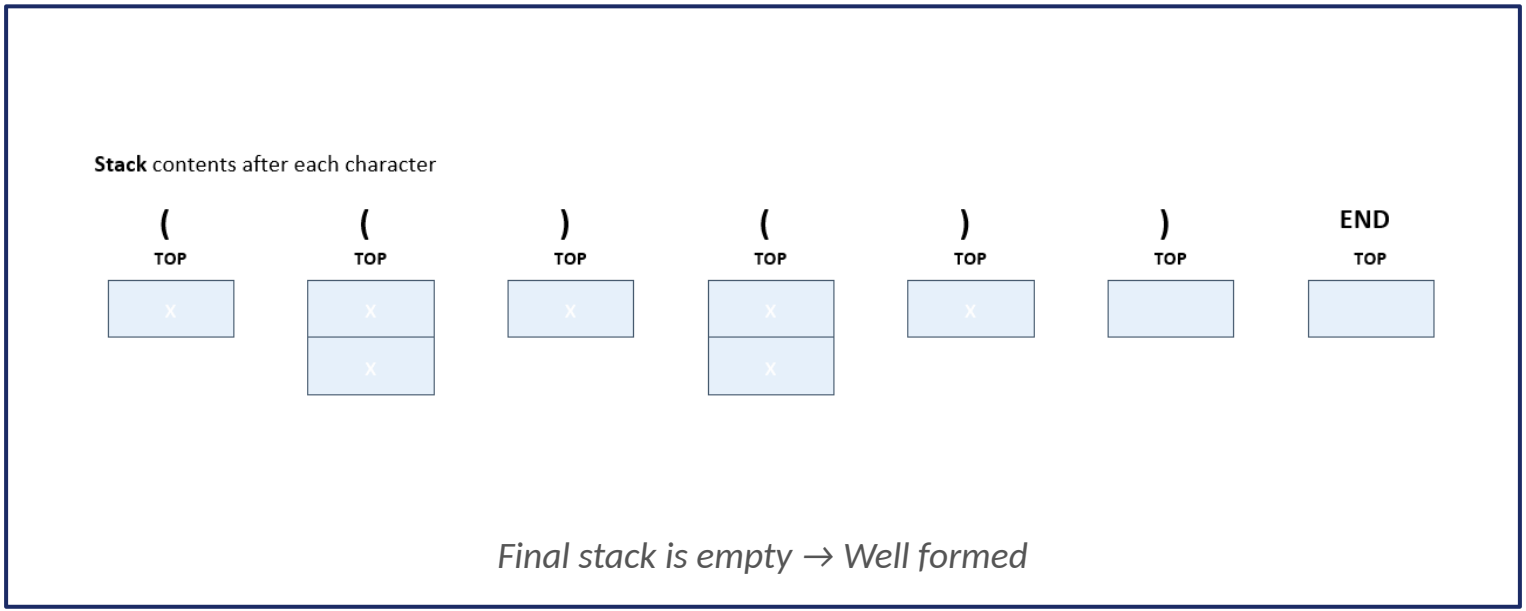
Stack : Applications

Algorithm: Well-Formed Parentheses

1. Initialize the stack.
2. Read the expression as a string and scan character by character.
3. If the character is '(', push a placeholder.
4. If the character is ')', pop; if the stack is empty, the expression is invalid.
5. After processing the expression, the stack must be empty.

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Example: Checking (()())



Final stack is empty → Well formed

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Algorithm: Well-Formed Parentheses : C Code

```
int is_valid = 1;
char ch; char *s = str;
while (is_valid && (ch = *s++) != '\0') {
    if (ch == '(')
        push(&mystack, 'x');
    else if (ch == ')') {
        if (!is_empty(&mystack))
            pop(&mystack);
        else
            is_valid = 0;
    }
}
```

```
if (is_valid && is_empty(&mystack))
    printf("Well formed expression");
else
    printf("not well formed expression");
```


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Algorithm: Well-Formed Brackets: (), [], {}

- With multiple bracket types, a generic placeholder is not enough.
- For '(' push ')'; for '[' push ']'; for '{' push '}'.
- When a closing bracket is encountered, the stack must not be empty.
- The closing bracket must match the character at the top of the stack.
- Well formed: (){}[], ([]{}) and ([{}]).
- Not well formed: ([)], ([]]) and (][).

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THANK YOU

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